

Project Design Phase

Proposed Solution Template

Date	18 June 2026
Team ID	
Project Name	ShopEz (MERN Full-Stack Application)
Student Name	SIDDANI SRI CHARAN (charaniamgreat@gmail.com)
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Existing e-commerce platforms often suffer from high latency, non-responsive mobile interfaces, and complex administrative workflows, leading to poor customer retention and inefficient inventory management.
2.	Idea / Solution description	ShopEz utilizes the MERN stack to deliver a high-performance, single-page application. It features real-time search, persistent cart management, intuitive category filtering, and a unified dashboard for streamlined order tracking and administrative management.
3.	Novelty / Uniqueness	• Full JavaScript implementation using MERN for seamless data flow. • Integrated Cloudinary-backed image management system. • Responsive design optimized for Men, Women, and Kids segments. • Granular admin controls for end-to-end order processing.
4.	Social Impact / Customer Satisfaction	Enhances digital accessibility by providing a secure and simplified shopping journey. Achieves customer satisfaction through minimal checkout steps, real-time inventory visibility, and accessible order history.
5.	Business Model (Revenue Model)	• Direct product sales and transaction fees. • Subscription-based premium features for vendors. • Targeted promotional listing fees.
6.	Scalability of the Solution	The MongoDB NoSQL architecture supports rapid data horizontal scaling. Future enhancements include multi-vendor support, integrated AI-driven recommendations, and multi-currency payment gateway expansion.